

Module 5: Computing (Containers and Serverless)

At the end of this module, you will be able to:

- **Explain** the purpose and function of AWS Lambda.
- **Describe** the purpose and function of application programming interfaces (APIs), including RESTful APIs.
- **Explain** the benefits and function of Amazon API Gateway.
- **Explain** the purpose and function of containers and the AWS services that support container usage.
- **Explain** the purpose and function of AWS Step Functions.

Topics:

- AWS Lambda
- APIs and REST
- Amazon API Gateway
- Containers on AWS
- AWS Step Functions

Knowledge check

Instructor Demo:

Deploy Docker Containers on Amazon Elastic Container Service (Amazon ECS)

Lab: None

Exercises / Project:

- Activity 5: Create a Lambda Function to Email Daily Reports
- Project: Troubleshooting Playbook

Troubleshooting knowledge base – Module 5 focus

- Have your copy of the troubleshooting knowledge base template open for editing
- **Create new entries** as you encounter troubleshooting situations
- **Categories of focus** for this module:
 - Compute

Module 5: Computing (Containers and Serverless)

AWS Lambda

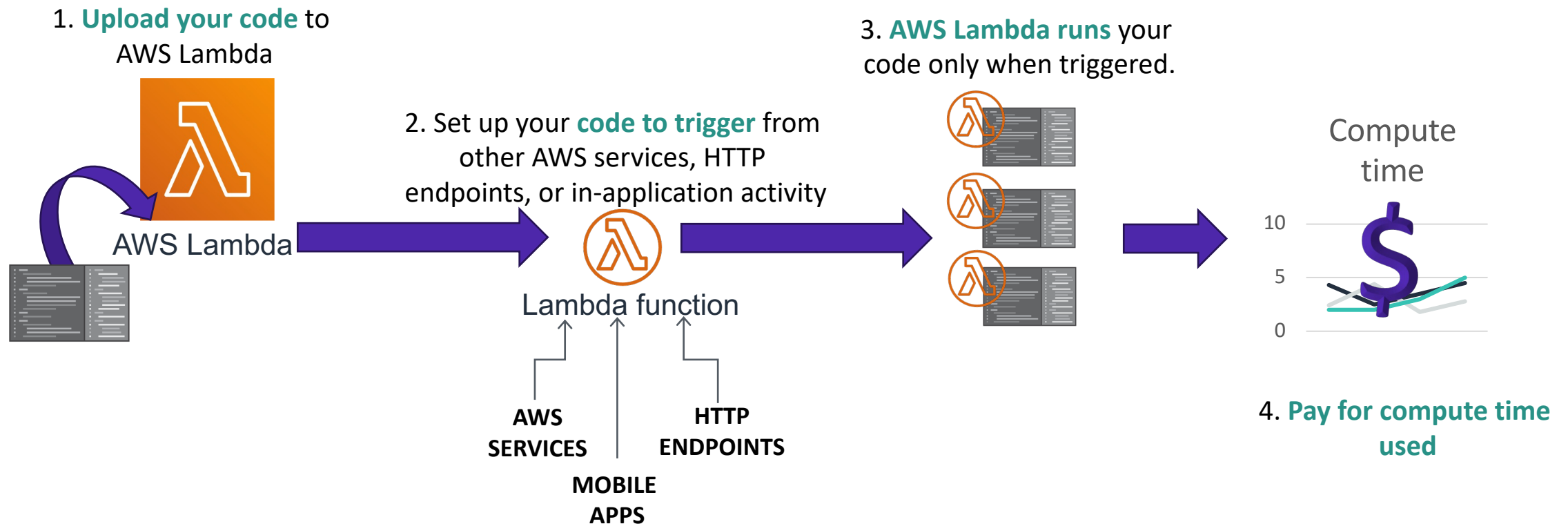
Traditional deployment and operations

- Provision an instance
- Update operating system (OS)
- Install application platform
- **Build and deploy applications**
- Configure automatic scaling and load balancing
- Regularly patch, secure, and monitor servers
- **Monitor and maintain applications**

Serverless deployment and operations

- Provision an instance
- Update operating system (OS)
- Install application platform
- **Build and deploy applications**
- Configure automatic scaling and load balancing
- Continuously patch, secure, and monitor servers
- **Monitor and maintain applications**

AWS Lambda allows code to run without provisioning or managing servers.



AWS Lambda key product features

AWS Lambda key product features include:

Bring your own code

Completely automated administration

Built-in fault tolerance

Automatic scaling

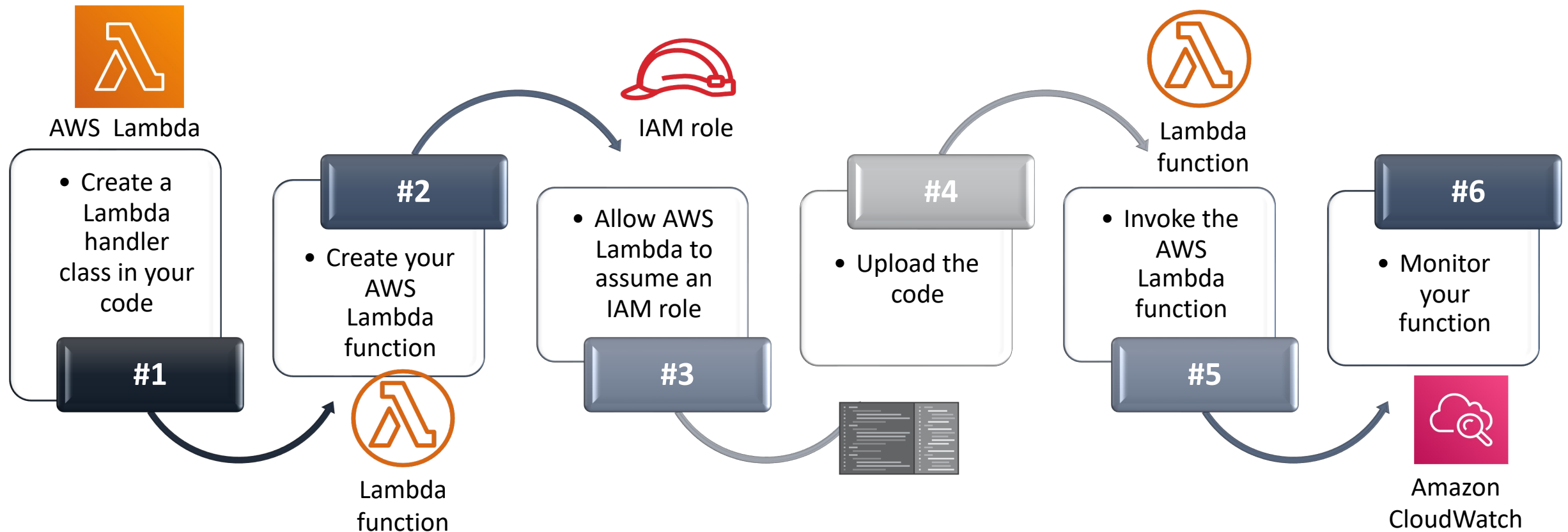
Orchestrate multiple functions

Integrated security model

Pay per use

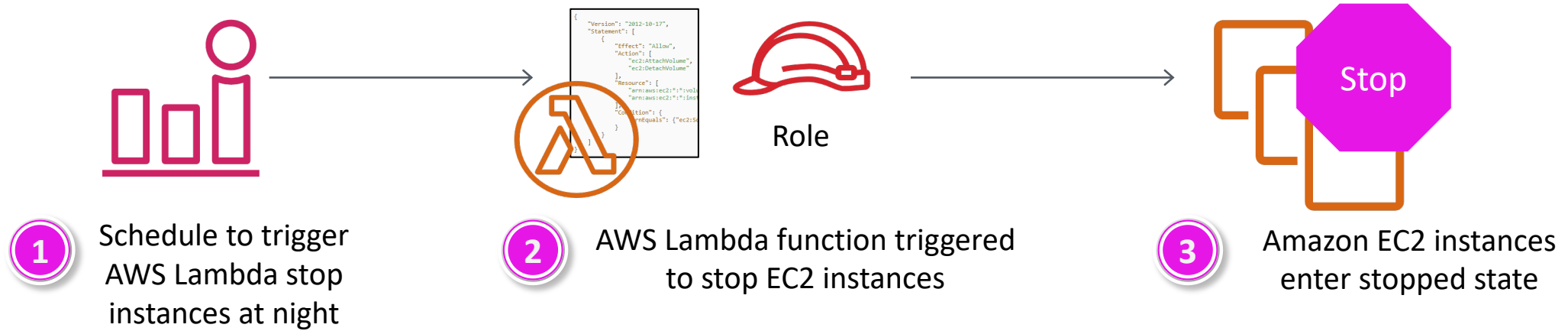
Flexible resource model

Steps to deploy AWS Lambda

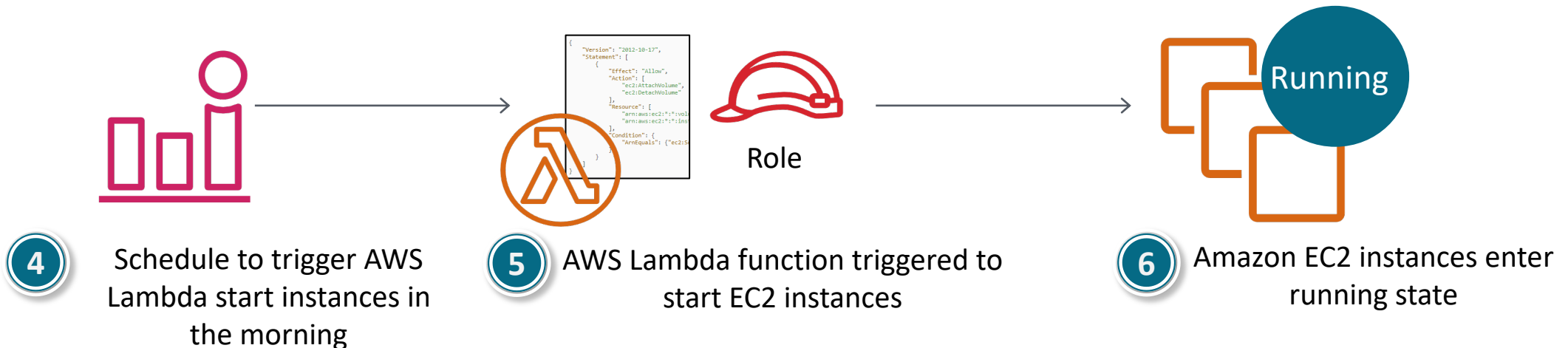


AWS Lambda example: Start and stop Amazon EC2 instances

Stop instances



Start instances



Lambda layers enable developers to:

- Configure a Lambda function to use libraries that are not included in the deployment package.
 - Lambda layer is a ZIP archive with libraries and a custom runtime
- Keep the deployment package small.
- Avoid package dependency errors in code.
- Share libraries with other developers.

AWS Lambda limits include:



AWS Lambda

Resource limits per invocation, such as memory, runtime, duration

Account limits per Region, such as concurrency

Deployment limits, such as package size

Functions that exceed any of the limits will fail with an **exceeded limits exception**

Key Takeaway



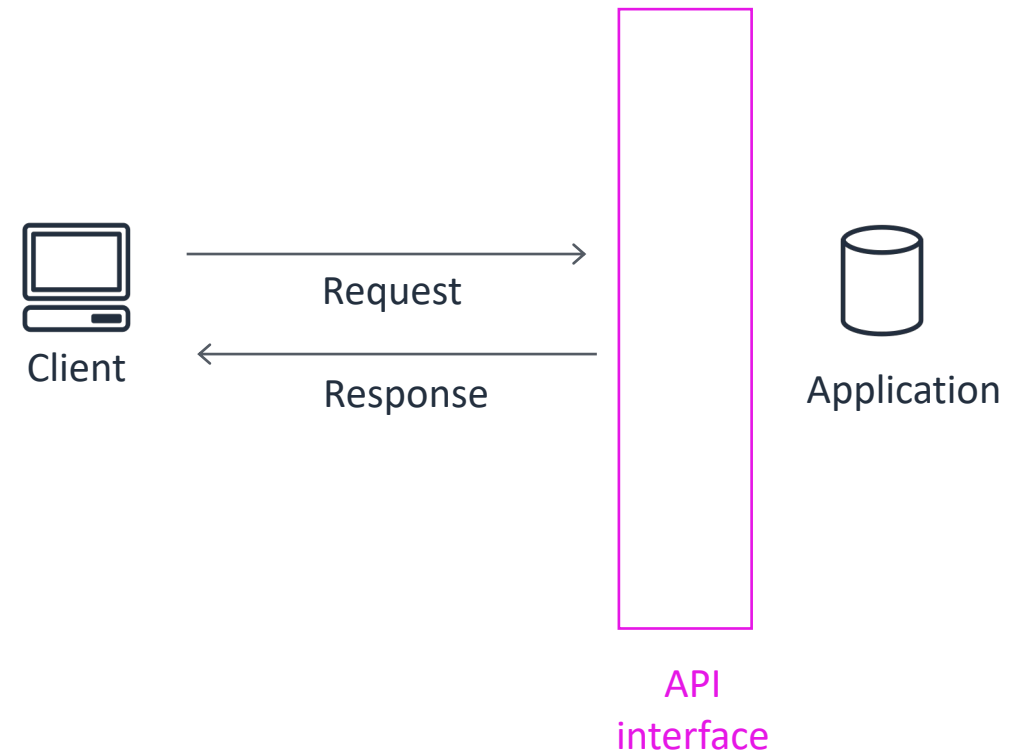
- Serverless computing allows you to build and run applications and services without provisioning or managing servers.
- AWS Lambda provides built-in fault tolerance and automatic scaling.
- The maximum memory allocation for a single Lambda function is 3 GB.
- The maximum runtime for a Lambda function is 15 minutes.

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APIs and REST

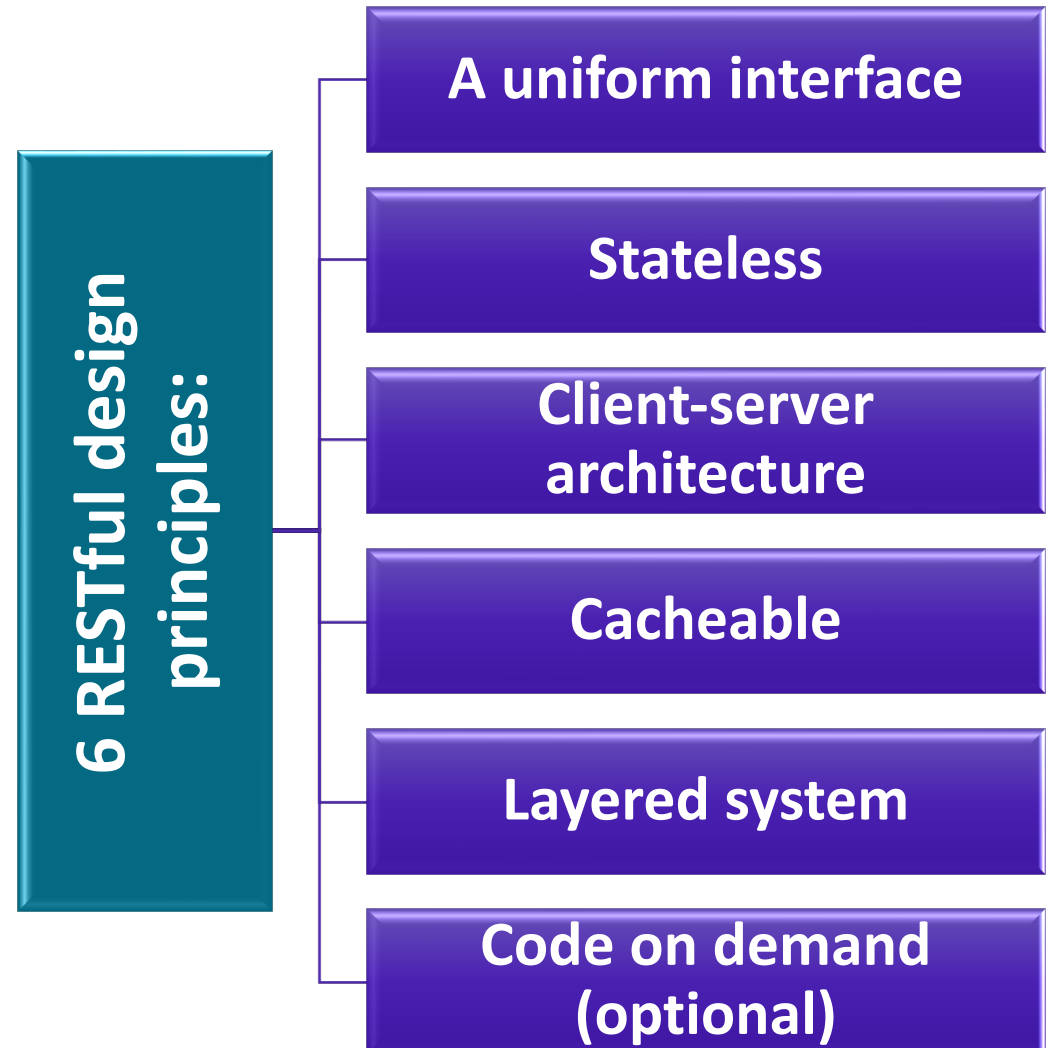
An **application programming interface (API)**:

- Provides programmatic access to an application. For example, they are often used for programs that communicate with each other.
- Supports multiple programming languages. For example, API for Java, API for C++, and so forth.



Representational state transfer (REST):

- A design for loosely coupled network-based applications
- Communication is over HTTP
- Exposes resources at specific URLs
- Many web APIs are RESTful



Components of **REST requests** include:

- Endpoint (in the form of a URL)
- Method:
 - **GET** – To read a resource
 - **POST** – To create a resource
 - **PUT** – To update an existing resource
 - **DELETE** – To delete a resource
- Header:
 - Example: -H "Content-Type: application/json"
- Data (body):
 - Typically part of POST and PUT requests

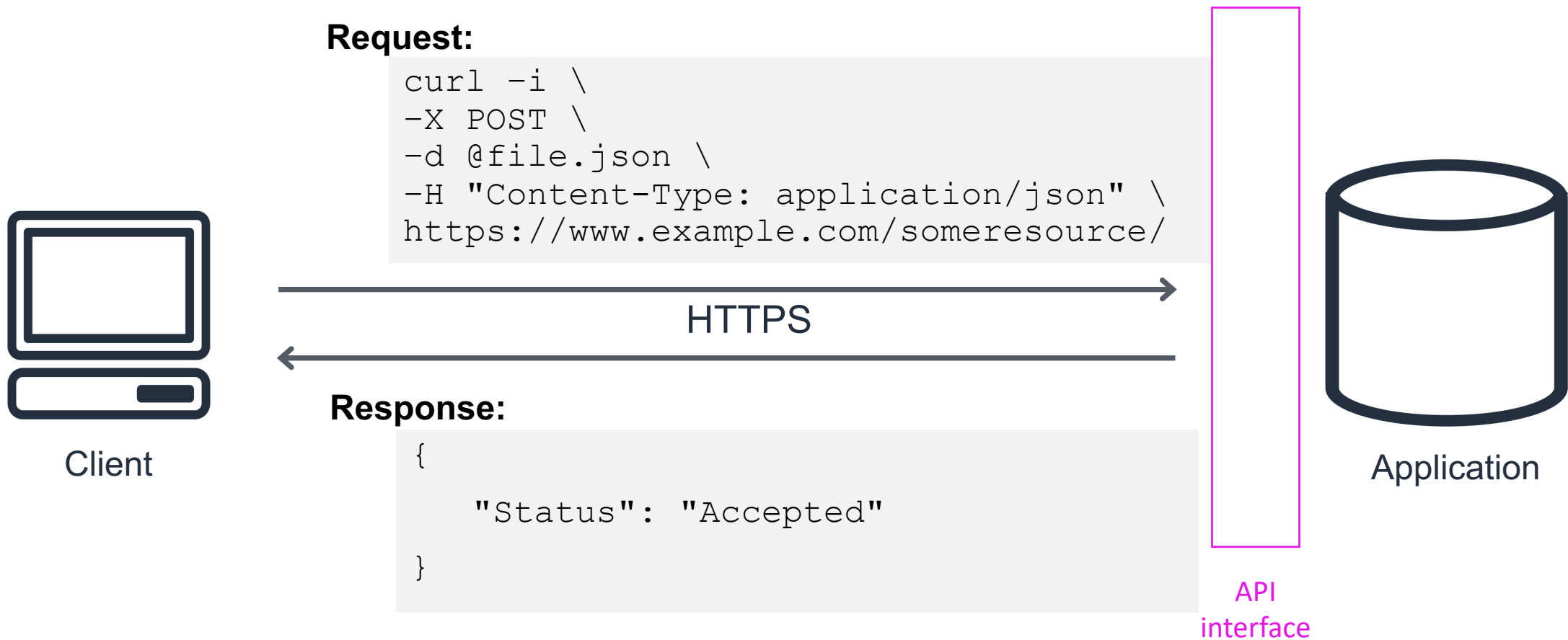
URL

Method

Headers

Body

REST example with cURL



HTTP status codes:

100s – Informational response

200s – Success

300s – Redirection

400s – Client errors

500s – Server errors

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Amazon API Gateway

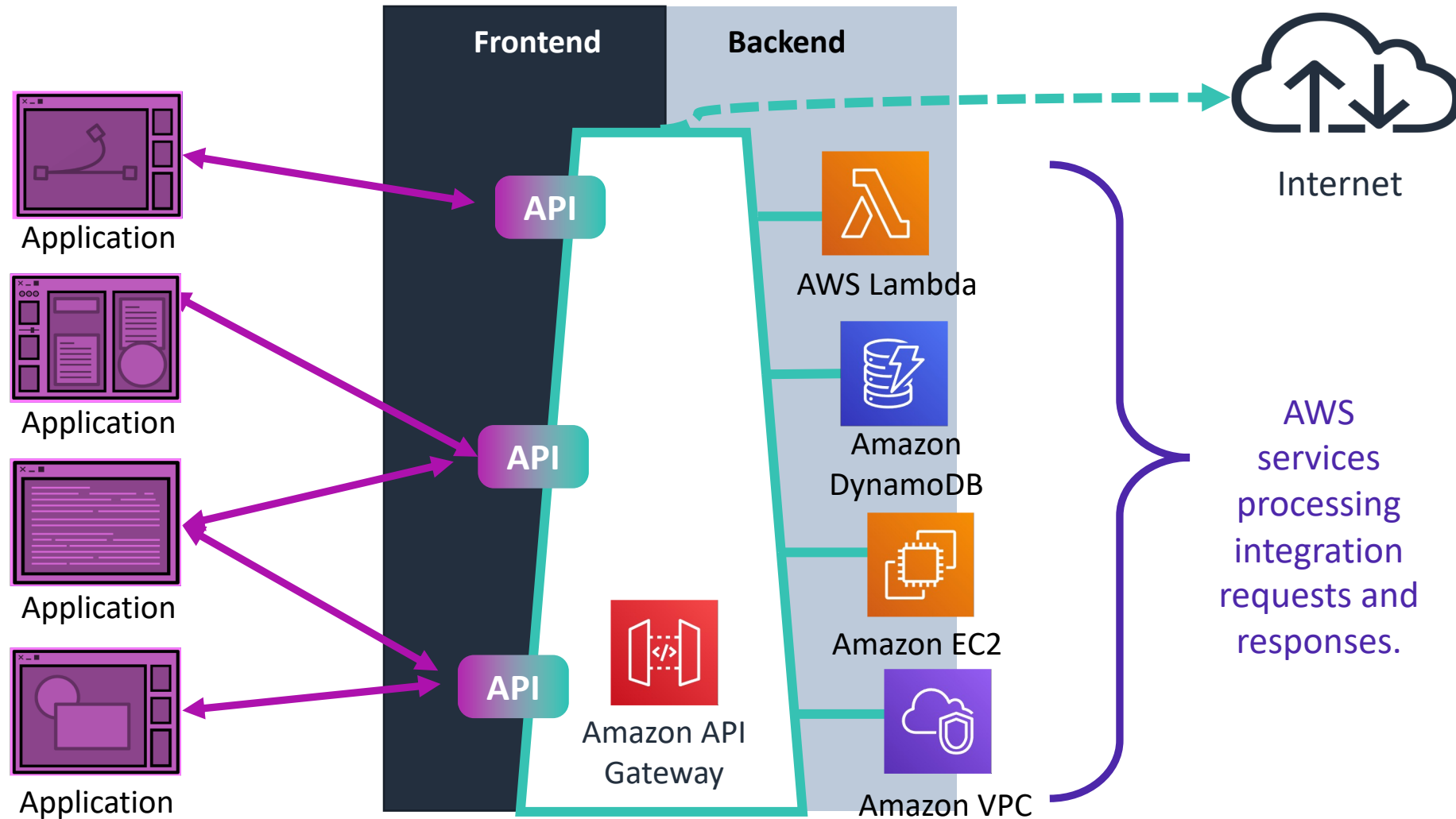
An Amazon API Gateway:



Amazon API Gateway

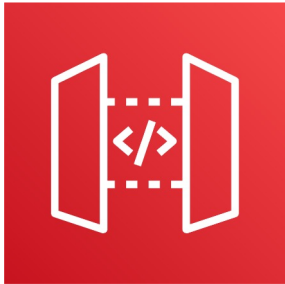
- Allows developers to create and maintain APIs. For example, create a REST API for an application you run on AWS.
- Is a fully managed service, which handles:
 - Scaling
 - Access control
 - Monitoring

Amazon API Gateway deployment



API Gateway benefits

Benefits of **API Gateway** include:



Amazon API Gateway

Benefit from low costs and efficiency

Provide performance at any scale

Easily monitor API activity

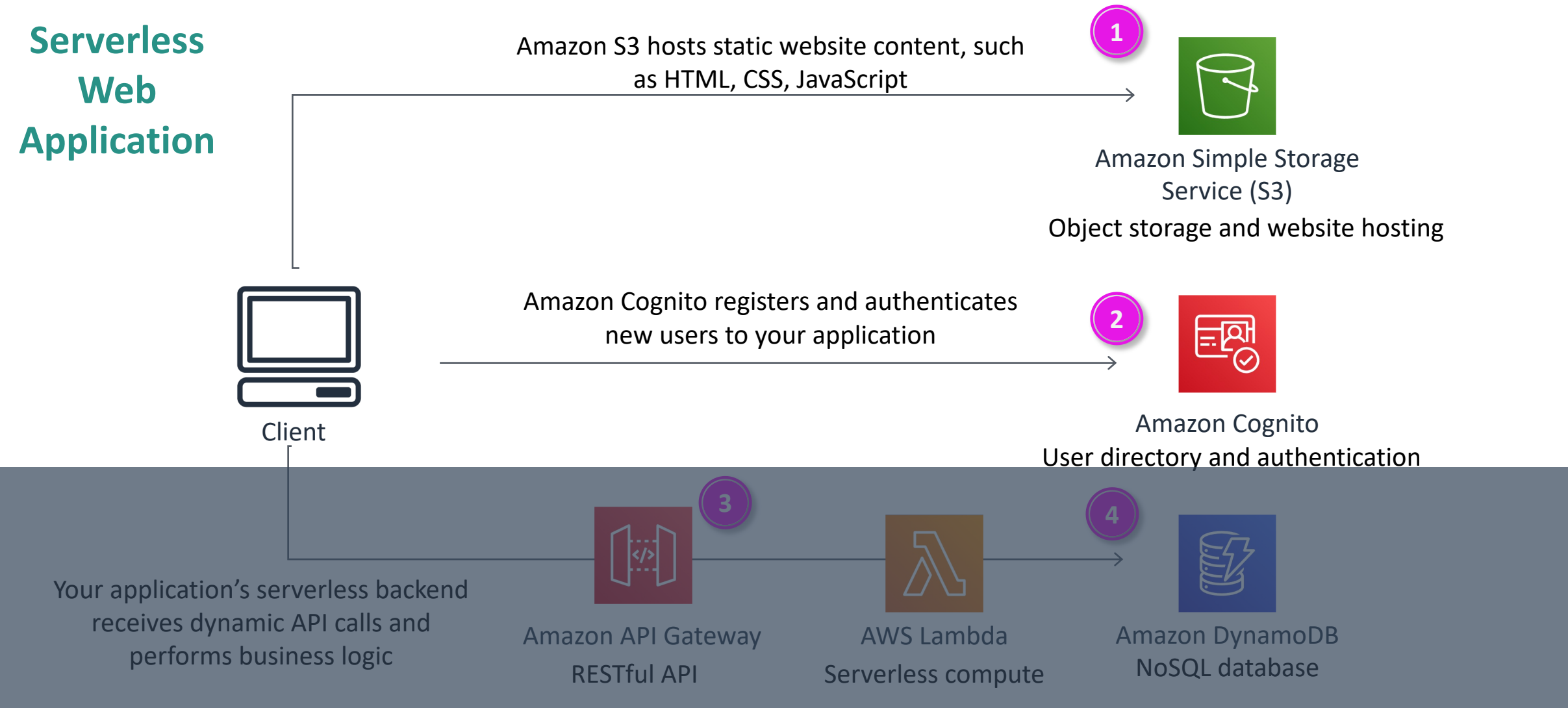
Streamline API development

Use flexible security controls

Run APIs without servers

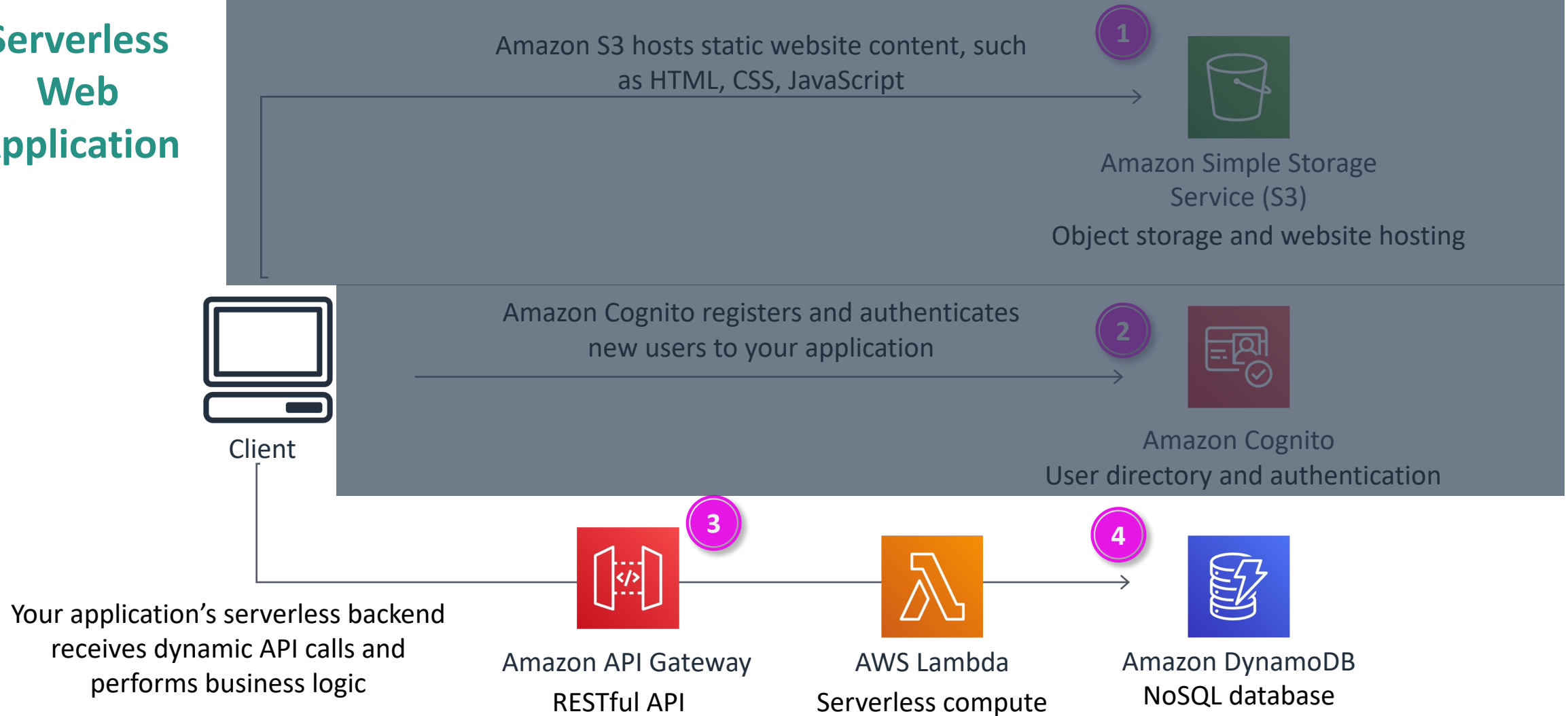
Create RESTful API endpoints for existing services

API Gateway example (1 of 2)



API Gateway example (2 of 2)

Serverless Web Application



Module 5: Computing (Containers and Serverless)

Containers on AWS

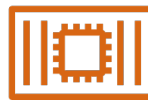
Introduction to containers

Containers are:

- A method of operating system virtualization.
- An application and its dependencies that can be run in resource-isolated processes.



Container 1



Container 2



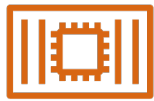
Container 3

Benefits of using containers

Containers:



Container 1



Container 2



Container 3

Environment consistency

Operational efficiency

Developer productivity

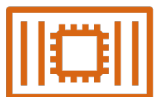
Version control

Docker is:

- A software platform that allows you to build, test, and deploy applications quickly.
- Used to package software into standardized units called *containers*.



Container 1



Container 2



Container 3

Containers have everything the software needs to run:

Libraries

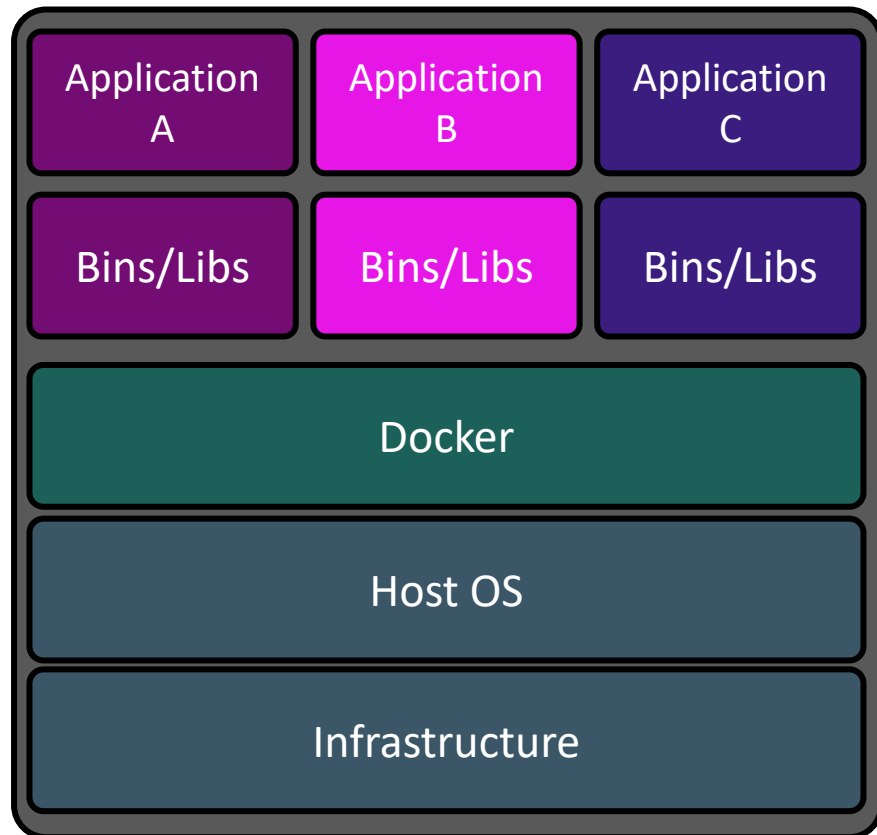
System
tools

Code

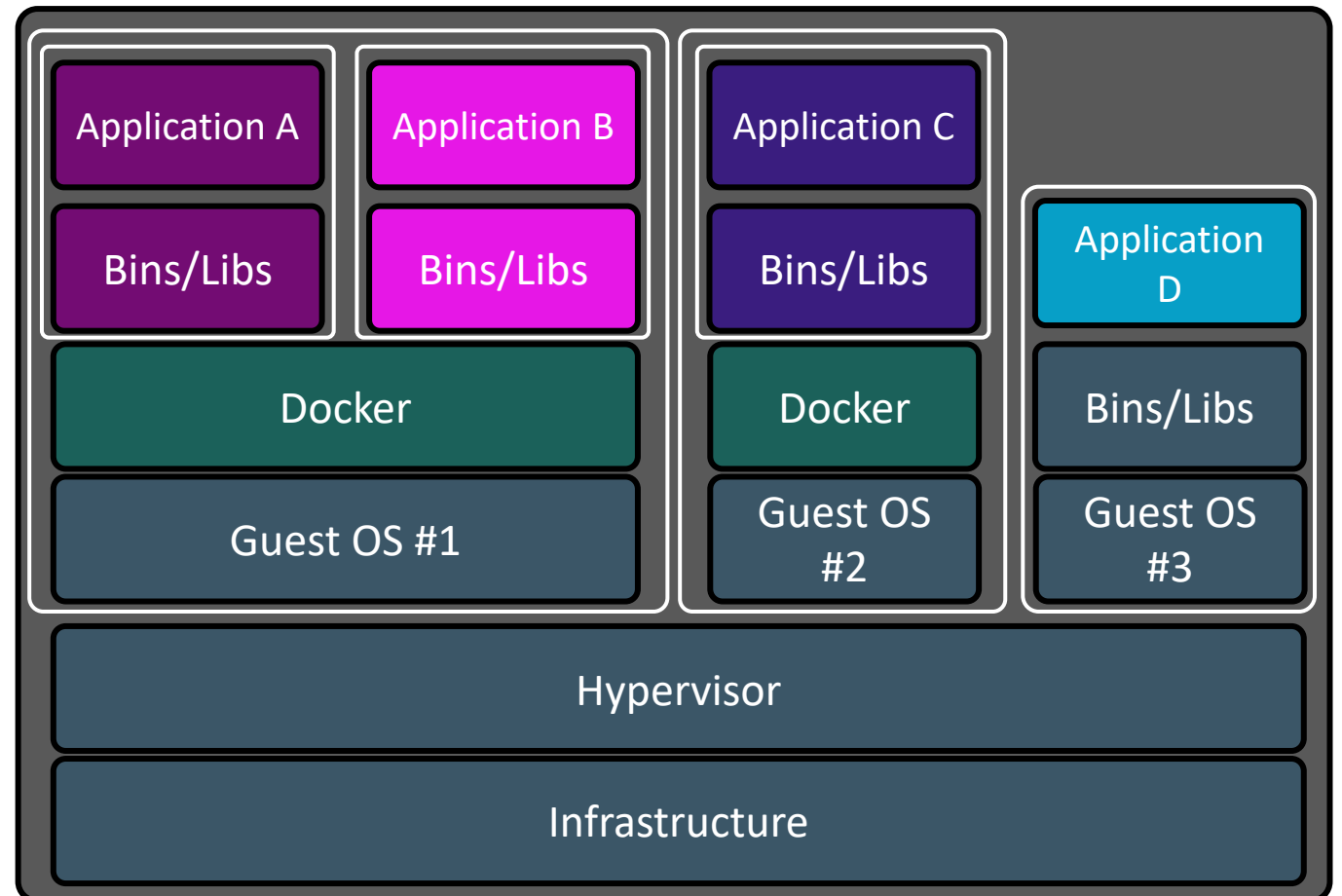
Runtime

Docker example deployments

Single host operating system, with Docker installed and how Docker would **run on premises**.



Four applications are hosted across three guest operating systems and how it would **run on AWS**.



Container management challenges (1 of 2)

Challenges of **Container management** include:

Container use is growing rapidly

Container sprawl avoidance

Version differences

Ownership

Container management challenges (2 of 2)

Challenges of **Container management** include:

Bin packing

Containers that are no longer needed (zombie containers)

Disappearing containers

Amazon Elastic Container Service (Amazon ECS)



Amazon Elastic
Container Service



Container 1



Container 2



Container 3



Task



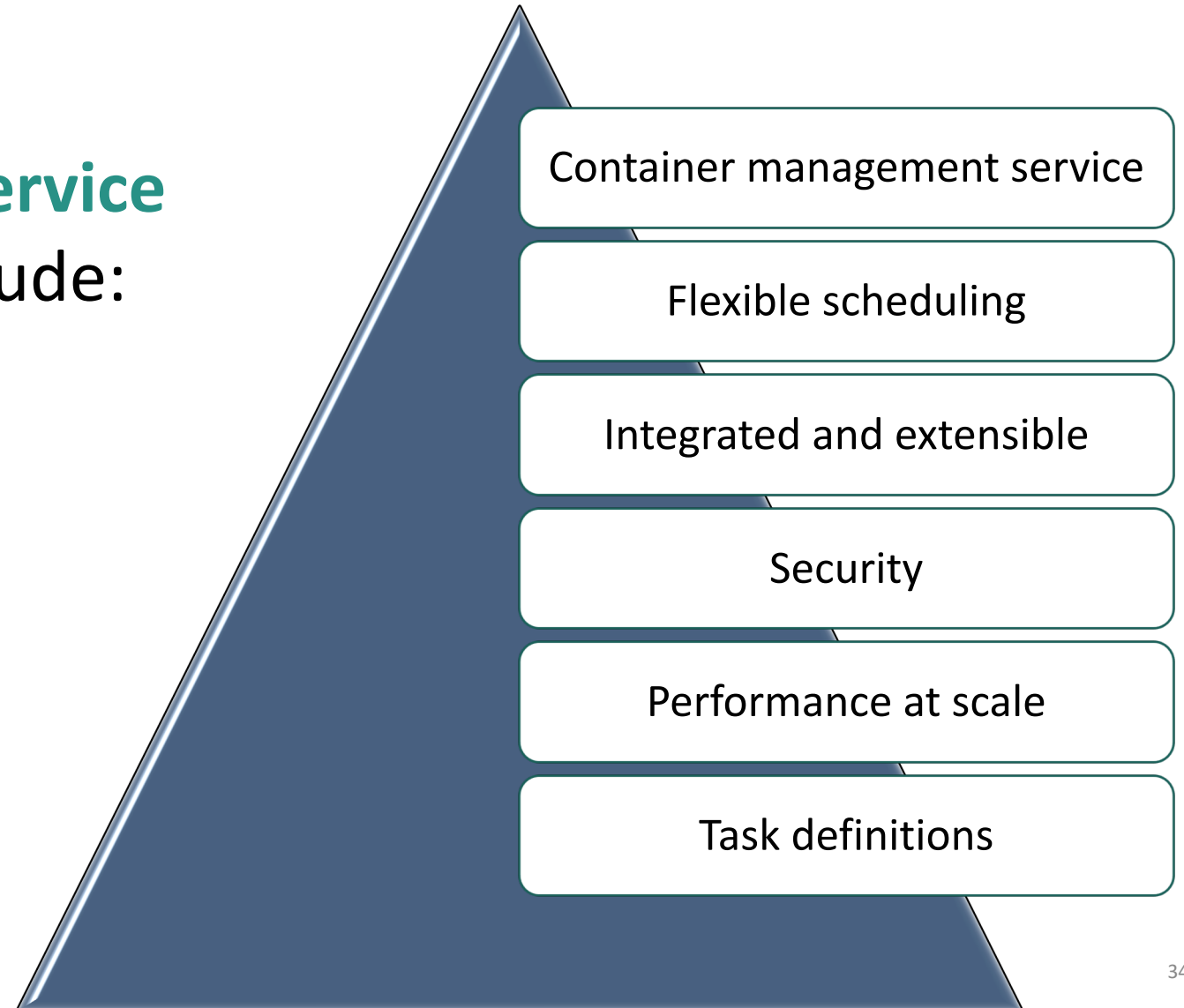
Service

Amazon Elastic Container Service (Amazon ECS) is:

- A highly scalable, high performance *container management service*
- Supports Docker containers.

Amazon Elastic Container Service (Amazon ECS) benefits

Amazon Elastic Container Service (Amazon ECS) benefits include:

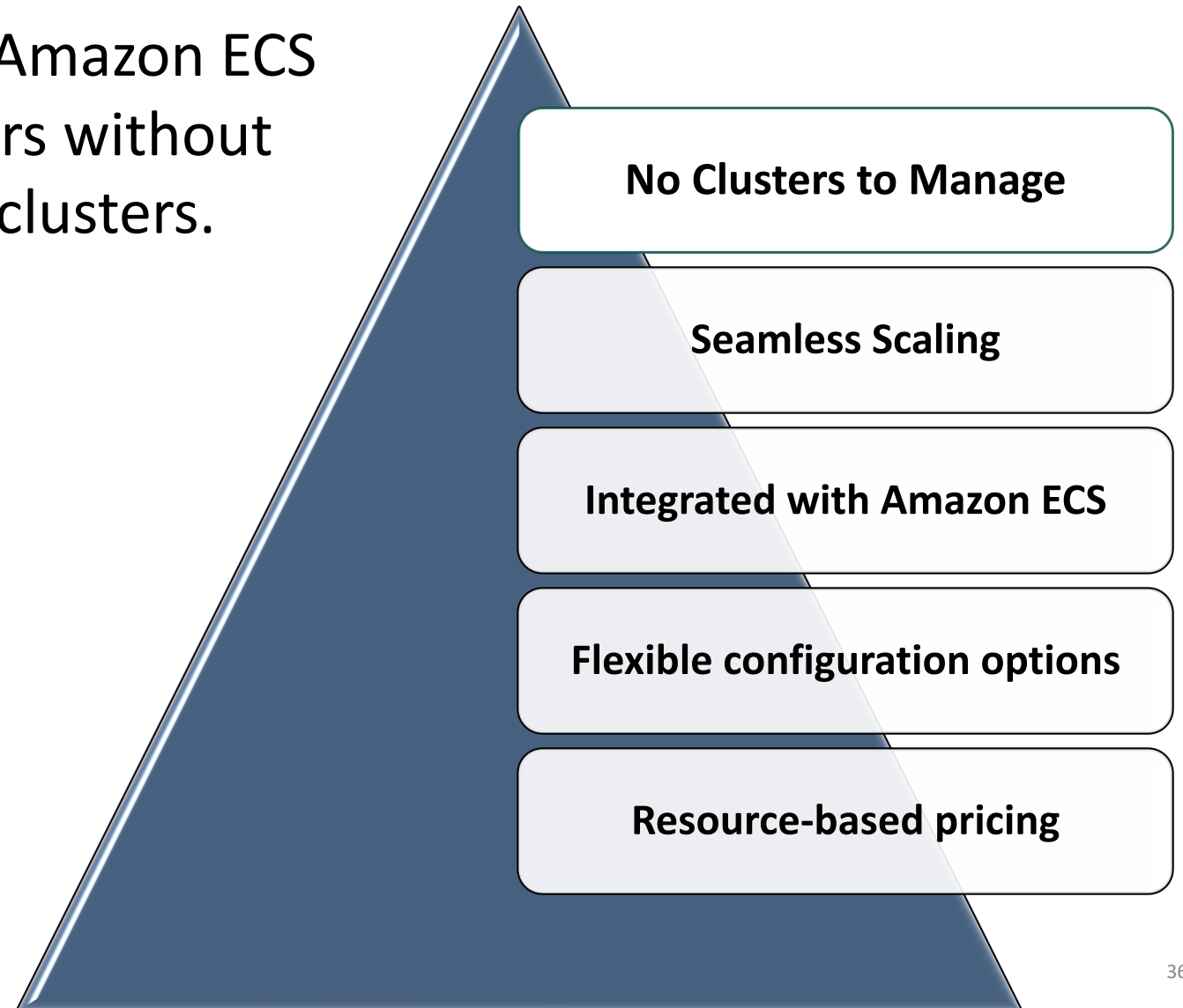


Demo: Deploy Docker container on Amazon Elastic Container Service (Amazon ECS)

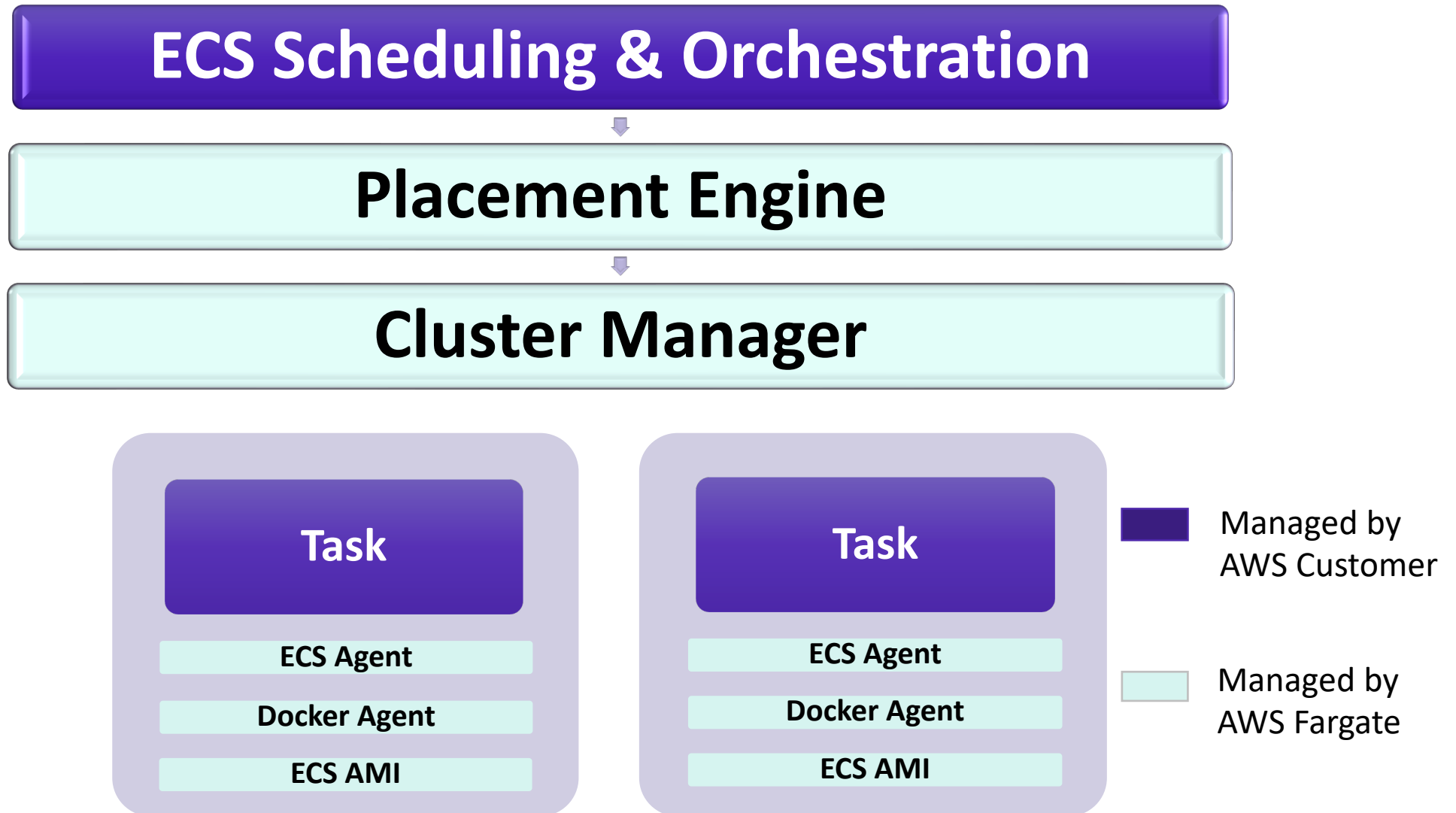
AWS Fargate is a technology for Amazon ECS that allows you to run containers without having to manage servers or clusters.



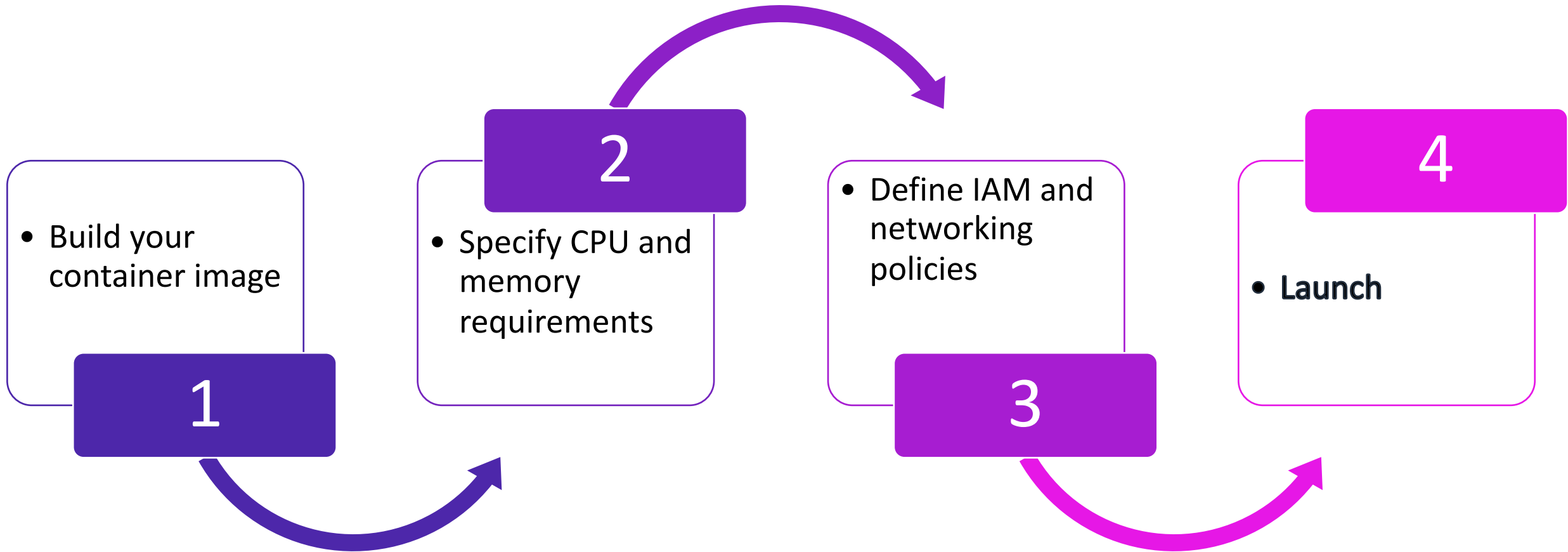
AWS Fargate



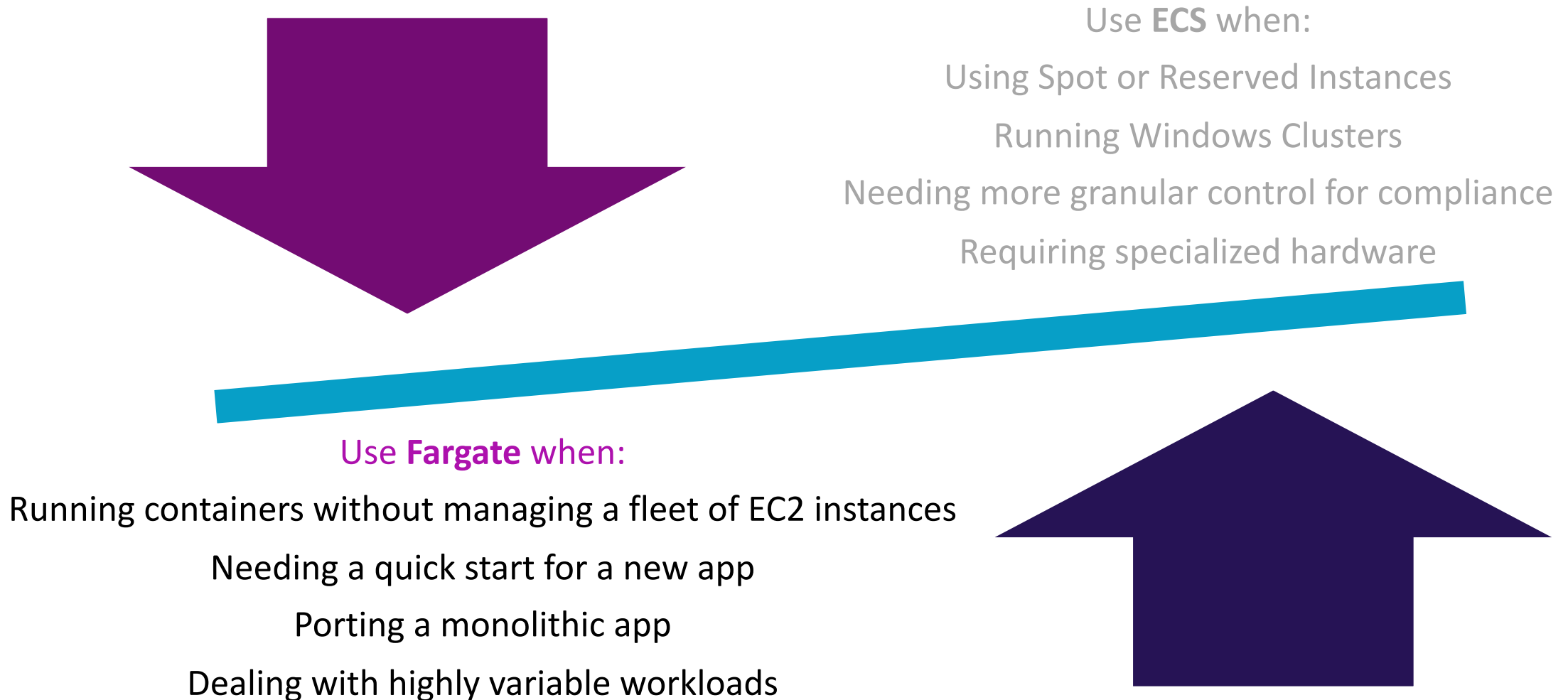
AWS Fargate and the ECS stack



Using AWS Fargate



When to use AWS Fargate over ECS



When to use ECS over AWS Fargate

Use **ECS** when:

Using Spot or Reserved Instances

Running Windows Clusters

Needing more granular control for compliance

Requiring specialized hardware

Use **Fargate** when:

Running containers without managing a fleet of EC2 instances

Needing a quick start for a new app

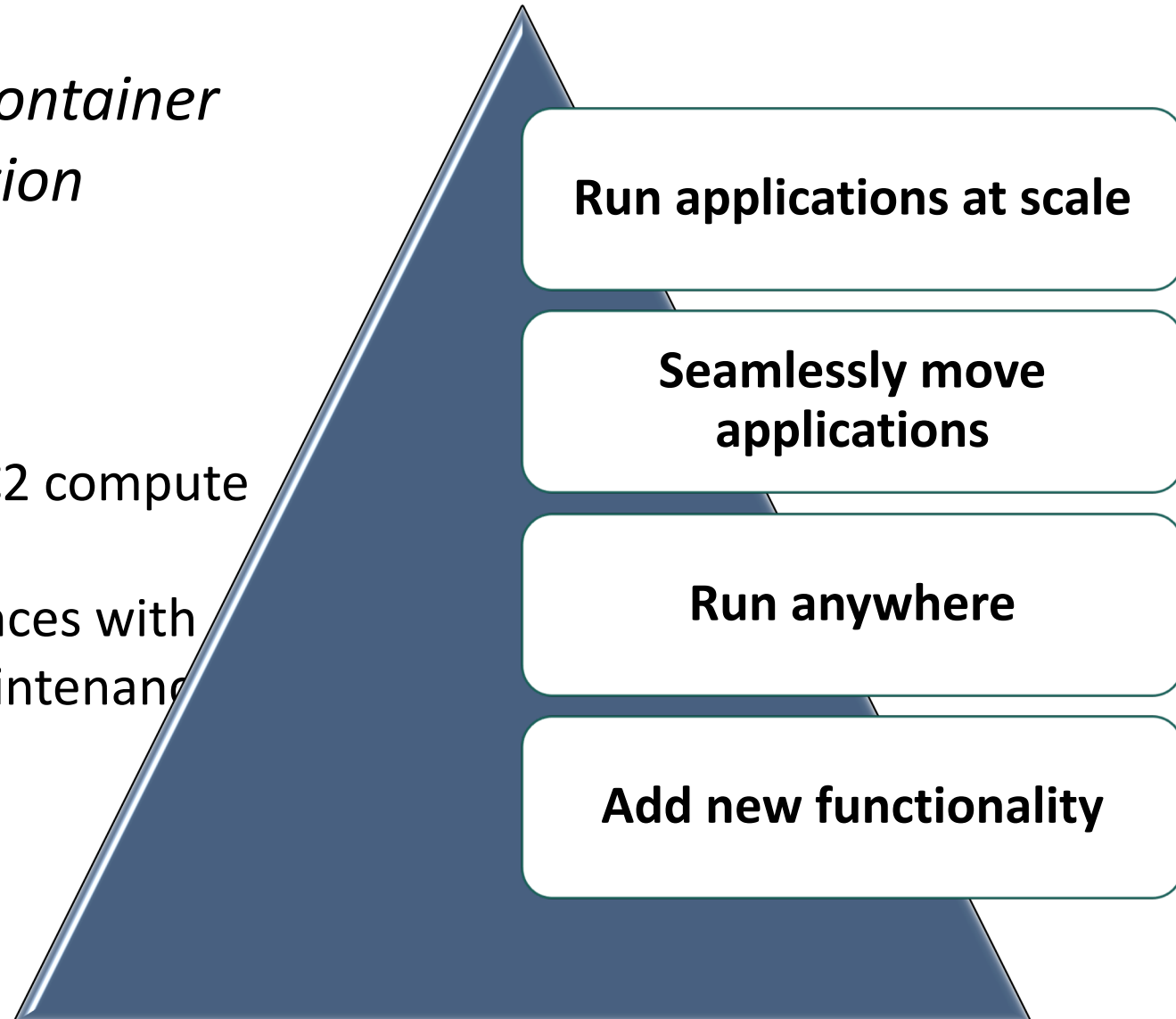
Porting a monolithic app

Dealing with highly variable workloads

Kubernetes is open source *container management and orchestration* software.

Use Kubernetes to:

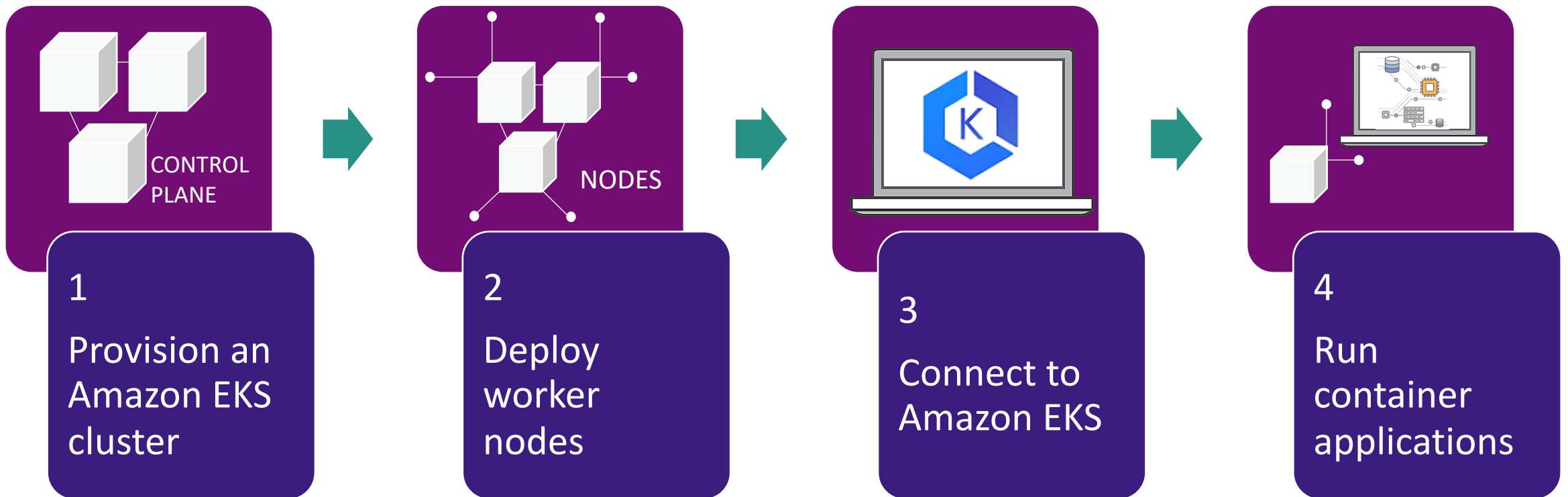
- Manage clusters of Amazon EC2 compute instances
- Run containers on those instances with processes for deployment, maintenance and scaling



Amazon Elastic Container Service for Kubernetes (Amazon EKS)



Amazon Elastic Container Service
for Kubernetes



Amazon Elastic Container Registry (Amazon ECR)

Amazon ECR is a fully managed Docker container registry that makes it easy for developers to store, manage, and deploy Docker container images.



**Amazon elastic container
service integration**

Docker support

High availability and durability

Team collaboration

Access control

Encryption at rest

Third-party integrations

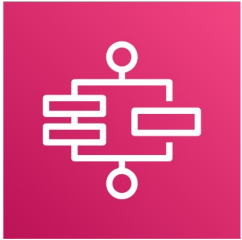
Key Takeaway



- Containers hold everything that the software needs to run.
- Docker is a software platform that packages software into containers.
- A single application can span multiple containers.
- Kubernetes is an open source container orchestration software.

Module 5: Computing (Containers and Serverless)

AWS Step Functions



AWS Step Functions



AWS Step Functions allow the ability to coordinate AWS services into serverless *workflows*.

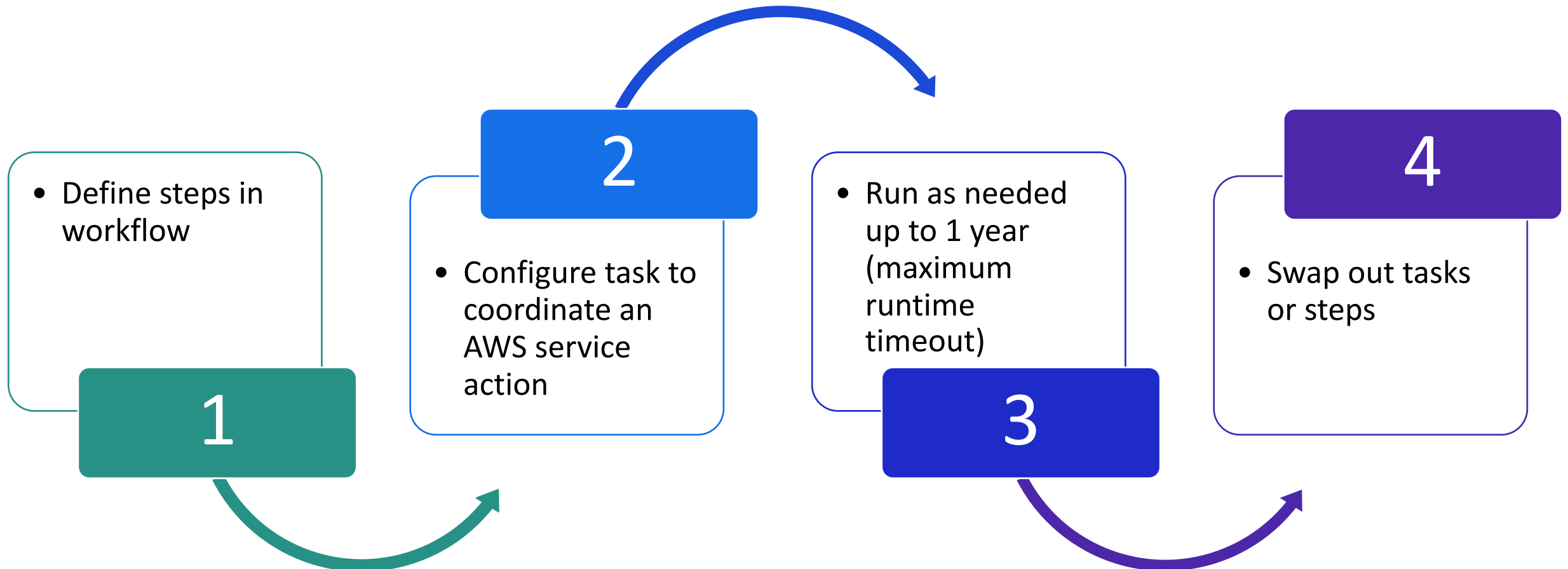
- Workflows consist of a series of steps.
- The output of one step is the input to the next step.
- Example:
 - AWS Step Functions enables AWS Lambda and Amazon Elastic Container Service (Amazon ECS) to work together

Benefits:

- Build and update distributed applications.
- Write less code.

How AWS Step Functions works

These steps show how **AWS Step Functions** works:



- The purpose and function of AWS Lambda.
- The purpose and function of application programming interfaces (APIs), including RESTful APIs.
- The benefits and function of Amazon API Gateway.
- The purpose and function of containers and the AWS services that support container usage.
- The purpose and function of AWS Step Functions.

To finish this module:

Complete:



Knowledge check

Test Strategies



Module 5 Scenario:

A trucking company is looking to minimize dependence on their current cloud service provider so they are not "locked in", while increasing their development velocity of various applications. The company is trying to decide if they should use containers or go serverless. They know they only need secure, compute capacity when the trucks are at their home base, which may last a day a couple times a month. Sometimes, they support partner companies when they are in town; therefore, they might need to accommodate those users once every couple of months.

Which statements describe when the company might find that containers are a more ideal option?
Select three responses.

A large, solid dark blue rectangular area intended for the user to select three responses to the question above.

Module 5 Scenario:

A trucking company is looking to minimize dependence on their current cloud service provider so they are not "locked in", while increasing their development velocity of various applications. The company is trying to decide if they should use containers or go serverless. They know they only need secure, compute capacity when the trucks are at their home base, which may last a day a couple times a month. Sometimes, they support partner companies when they are in town; therefore, they might need to accommodate those users once every couple of months.

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Module 5 Scenario:

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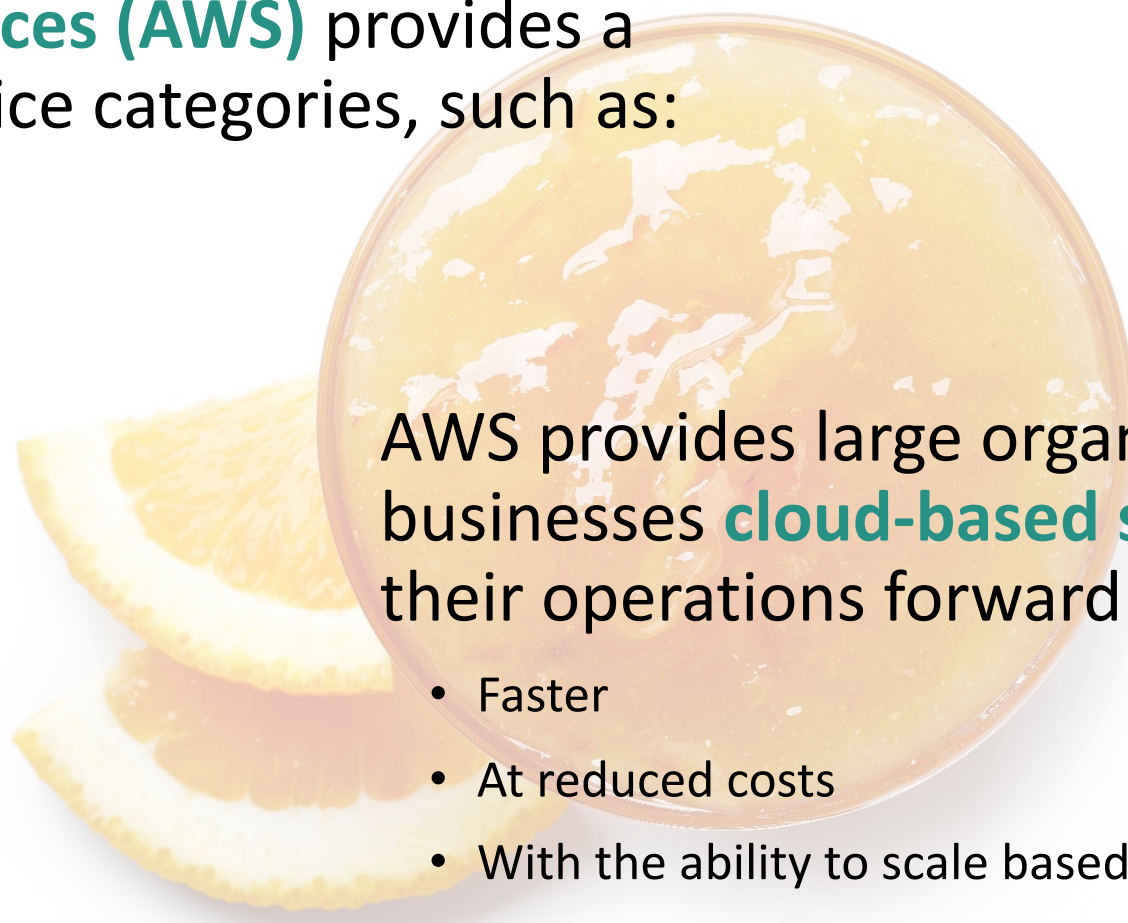
- A. Have the ability to scale back or tear down as needed
- B. Desire to manage the resources and their respective policies, and oversee security
- C. Ensure that the Cloud Service Provider provisions the infrastructure and manages capacity
- D. Move software from one environment to another through isolation
- E. Have a requirement for portability and maintaining a status of vendor-agnosticism

Systems Operations Project Scenario

Mom & Pop Café & Bakery

Amazon Web Services (AWS) provides a broad array of service categories, such as:

- Compute
- Storage
- Database
- Analytics

A background image featuring a glass of orange juice with a slice of orange floating in it, and two more orange slices on the surface below the glass.

AWS provides large organizations and small businesses **cloud-based solutions** to move their operations forward:

- Faster
- At reduced costs
- With the ability to scale based on demand

On completion of this project, you will be able to:

- **Experience** the communication challenges faced when attempting to apply technology as the solution to business problems.
- **Translate** customer requirements into a proposed technical solution.
- **Present** the proposed solution to the customer.
- **Use** AWS Best Practices for a frugal, resilient, and secure solution.



Mom & Pop Café & Bakery

Customer meeting preparation

Mom & Pop Café & Bakery

Request to send daily order history

Business need: Automated reporting



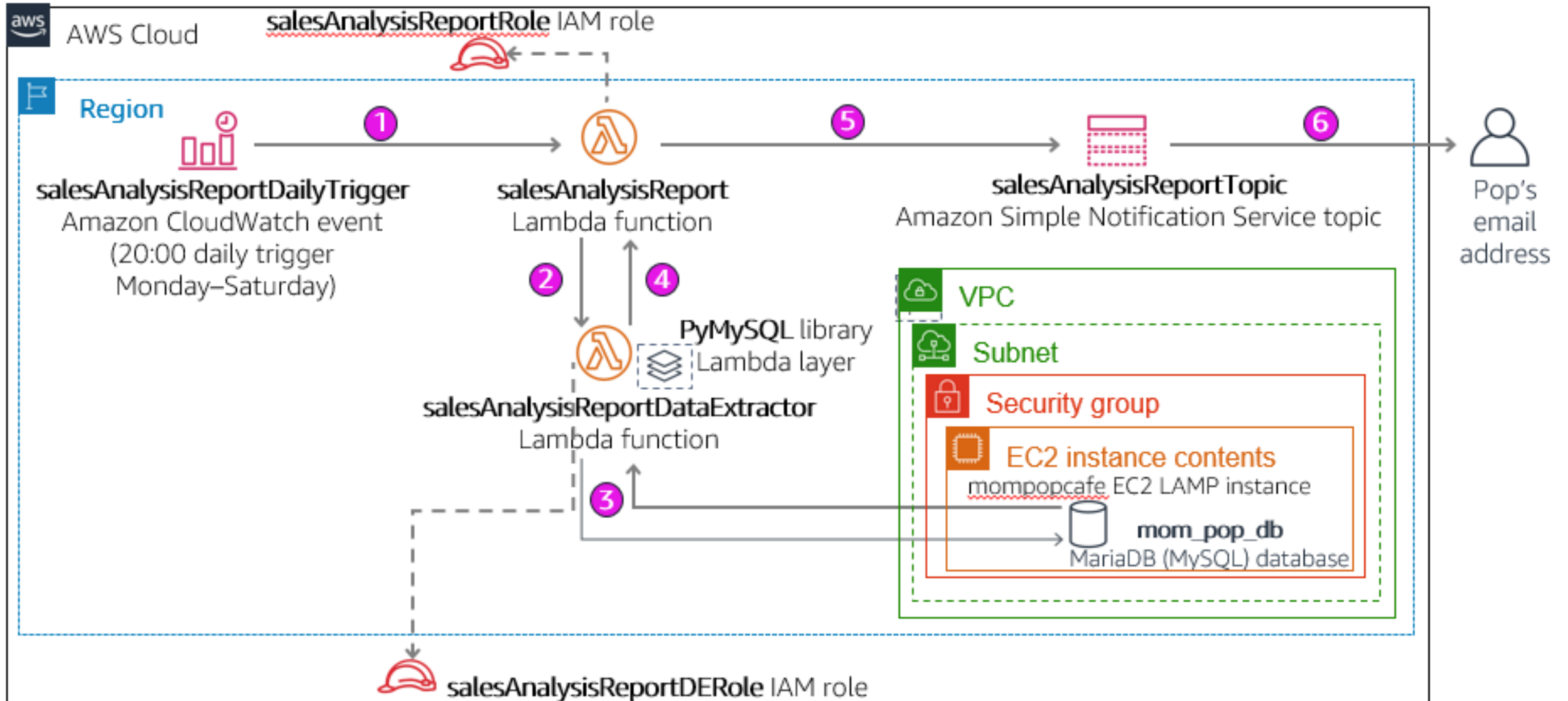
Sophie wants to send out a **daily report** to Mom and Pop. This report would list the café's daily baking needs.

The report would use previous orders to determine tomorrow's baking requirements.

Activity 5: Create an AWS Lambda function to email daily reports



Activity #5: Sales analysis report architecture and flow diagram





Begin Activity #5: Create Lambda function to send email reports

AWS Lambda

**What other business requirements
could AWS Lambda potentially solve
for the Mom & Pop Café?**

AWS Lambda

What other business requirements could AWS Lambda potentially solve for the Mom & Pop Café?

Solution

The AWS Lambda function could extract web access information from log files and send relevant information to Mom and Pop.

Project time: Troubleshooting knowledge base



Thank you

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